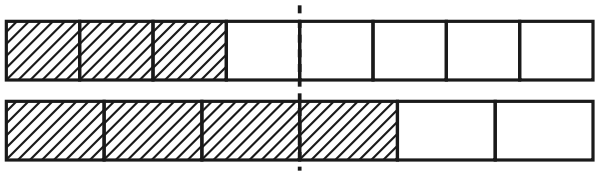
Comparing Unlike Fractions

Use 1/2, or make the pieces the same size.

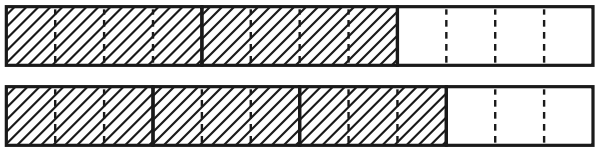
Compare with 1/2



$\frac{3}{8} < \frac{4}{6}$

3/8 is less than half.
4/6 is more than half.

Make the bottom numbers the same



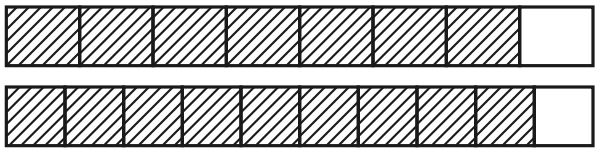
$\frac{2}{3} = \frac{8}{12}$

$\frac{3}{4} = \frac{9}{12}$

Now the pieces match:

$\frac{2}{3} < \frac{3}{4}$

Close to 1? Look at the missing piece



$\frac{7}{8} < \frac{9}{10}$

7/8 misses 1/8.
9/10 misses only 1/10.

Your turn!

$\frac{2}{5}$ [<] $\frac{5}{8}$

Which one is more than half?

For teachers

Goal: compare fractions with different tops and bottoms (4.NF.2). Ask “Is each one more or less than half?” If that doesn't settle it, cut the pieces so the bottoms match.

Comparing Unlike Fractions: Practice

Same whole. Write <, > or =.

Part 1 More or less than half? Half of the bottom number is the middle.

1) $\frac{3}{8}$ less than 1/2 = 1/2 more than 1/2

2) $\frac{4}{6}$ less than 1/2 = 1/2 more than 1/2

3) $\frac{5}{10}$ less than 1/2 = 1/2 more than 1/2

4) $\frac{2}{5}$ less than 1/2 = 1/2 more than 1/2

5) $\frac{7}{12}$ less than 1/2 = 1/2 more than 1/2

6) $\frac{3}{4}$ less than 1/2 = 1/2 more than 1/2

7) $\frac{6}{12}$ less than 1/2 = 1/2 more than 1/2

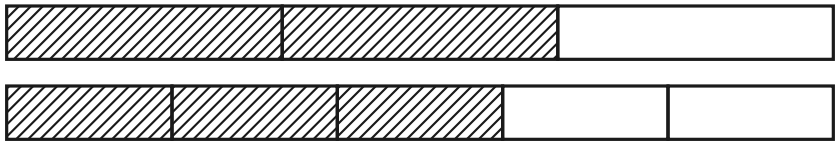
8) $\frac{1}{3}$ less than 1/2 = 1/2 more than 1/2

9) $\frac{4}{5}$ less than 1/2 = 1/2 more than 1/2

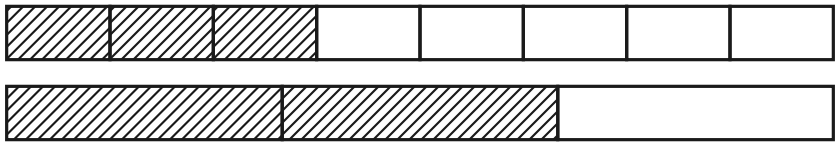
10) $\frac{5}{12}$ less than 1/2 = 1/2 more than 1/2

Comparing Unlike Fractions: Practice (continued)

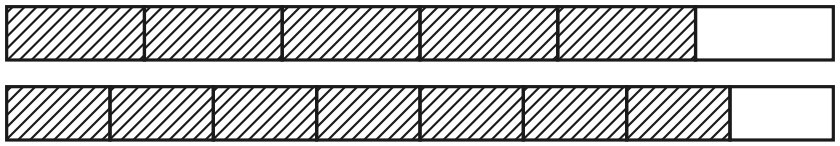
Part 2 Compare with strips Look at how much is shaded.

11) 

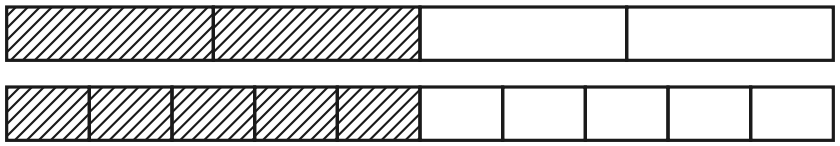
$\frac{2}{3} > \frac{3}{5}$

12) 

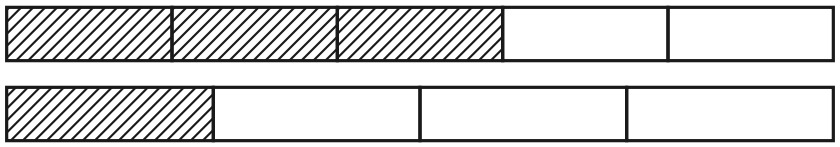
$\frac{3}{8} < \frac{2}{3}$

13) 

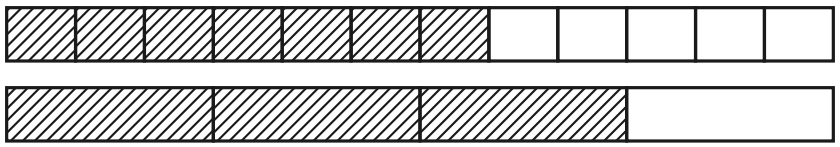
$\frac{5}{6} < \frac{7}{8}$

14) 

$\frac{2}{4} = \frac{5}{10}$

15) 

$\frac{3}{5} > \frac{1}{4}$

16) 

$\frac{7}{12} < \frac{3}{4}$

Comparing Unlike Fractions: Practice (continued)

Part 3 Use 1/2 No help this time. One is less than half, one is more.

17) $\frac{3}{8} < \frac{4}{6}$

18) $\frac{5}{6} > \frac{2}{5}$

19) $\frac{5}{8} > \frac{2}{6}$

20) $\frac{1}{3} < \frac{7}{10}$

21) $\frac{4}{10} < \frac{2}{3}$

22) $\frac{7}{12} > \frac{3}{8}$

23) $\frac{4}{6} > \frac{1}{4}$

24) $\frac{2}{5} < \frac{3}{4}$

25) $\frac{3}{10} < \frac{5}{8}$

Part 4 Make the bottoms the same Then compare the tops.

26) $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$ so $\frac{2}{3} < \frac{3}{4}$

27) $\frac{3}{5} = \frac{6}{10}$ and $\frac{7}{10} = \frac{7}{10}$ so $\frac{3}{5} < \frac{7}{10}$

28) $\frac{5}{6} = \frac{10}{12}$ and $\frac{3}{4} = \frac{9}{12}$ so $\frac{5}{6} > \frac{3}{4}$

Comparing Unlike Fractions: Practice (continued)

Part 4 Make the bottoms the same (continued)

29) $\frac{3}{8} = \frac{\boxed{3}}{8}$ and $\frac{1}{4} = \frac{\boxed{2}}{8}$ so $\frac{3}{8}$ $\boxed{>}$ $\frac{1}{4}$

30) $\frac{5}{12} = \frac{\boxed{5}}{12}$ and $\frac{1}{3} = \frac{\boxed{4}}{12}$ so $\frac{5}{12}$ $\boxed{>}$ $\frac{1}{3}$

31) $\frac{2}{5} = \frac{\boxed{4}}{10}$ and $\frac{3}{10} = \frac{\boxed{3}}{10}$ so $\frac{2}{5}$ $\boxed{>}$ $\frac{3}{10}$

32) $\frac{1}{2} = \frac{\boxed{6}}{12}$ and $\frac{5}{12} = \frac{\boxed{5}}{12}$ so $\frac{1}{2}$ $\boxed{>}$ $\frac{5}{12}$

33) $\frac{4}{6} = \frac{\boxed{8}}{12}$ and $\frac{7}{12} = \frac{\boxed{7}}{12}$ so $\frac{4}{6}$ $\boxed{>}$ $\frac{7}{12}$

Part 5 Close to 1 No help this time. Which one misses less?

34) $\frac{7}{8}$ $\boxed{<}$ $\frac{9}{10}$

35) $\frac{5}{6}$ $\boxed{>}$ $\frac{3}{4}$

36) $\frac{11}{12}$ $\boxed{>}$ $\frac{4}{5}$

37) $\frac{2}{3}$ $\boxed{<}$ $\frac{5}{6}$

38) $\frac{9}{10}$ $\boxed{>}$ $\frac{3}{4}$

39) $\frac{4}{5}$ $\boxed{<}$ $\frac{7}{8}$

Comparing Unlike Fractions: Practice (continued)

Part 6 Compare No help this time.

40) $\frac{3}{4} > \frac{5}{8}$

41) $\frac{7}{12} < \frac{5}{6}$

42) $\frac{4}{5} > \frac{7}{10}$

43) $\frac{5}{12} < \frac{1}{2}$

44) $\frac{3}{10} < \frac{2}{5}$

45) $\frac{1}{4} < \frac{3}{8}$

46) $\frac{5}{6} > \frac{7}{12}$

47) $\frac{6}{10} = \frac{3}{5}$

48) $\frac{7}{10} > \frac{1}{2}$

49) $\frac{1}{3} < \frac{5}{12}$

50) $\frac{2}{4} < \frac{7}{12}$

51) $\frac{3}{8} < \frac{2}{4}$

Comparing Unlike Fractions: Practice (continued)

Part 7 Circle the smallest No help this time.

52) $\frac{3}{4}$ $\frac{1}{3}$ $\frac{5}{8}$

53) $\frac{7}{10}$ $\frac{4}{5}$ $\frac{1}{2}$

54) $\frac{5}{6}$ $\frac{2}{3}$ $\frac{7}{12}$

55) $\frac{3}{5}$ $\frac{1}{2}$ $\frac{9}{10}$

56) $\frac{2}{6}$ $\frac{5}{8}$ $\frac{1}{4}$

57) $\frac{3}{8}$ $\frac{1}{2}$ $\frac{2}{3}$

Part 8 Compare Write <, > or =.

58) $\frac{2}{3} < \frac{10}{12}$

59) $\frac{5}{6} > \frac{5}{8}$

60) $\frac{6}{10} < \frac{4}{6}$

61) $\frac{1}{4} < \frac{11}{12}$

62) $\frac{5}{8} > \frac{2}{4}$

63) $\frac{2}{6} < \frac{4}{10}$

Comparing Unlike Fractions: Practice (continued)

Part 8 Compare (continued)

64) $\frac{3}{5} > \frac{1}{8}$

65) $\frac{7}{12} < \frac{6}{8}$

66) $\frac{3}{12} < \frac{3}{6}$

67) $\frac{6}{8} > \frac{4}{10}$

68) $\frac{2}{5} < \frac{7}{12}$

69) $\frac{7}{10} > \frac{1}{6}$

70) $\frac{2}{4} < \frac{9}{10}$

71) $\frac{10}{12} > \frac{2}{3}$

72) $\frac{2}{12} < \frac{2}{3}$

73) $\frac{3}{12} < \frac{9}{10}$

74) $\frac{4}{12} < \frac{5}{6}$

75) $\frac{7}{8} > \frac{10}{12}$

Comparing Unlike Fractions: Practice (continued)

Part 8 Compare (continued)

76) $\frac{2}{4} > \frac{3}{10}$

77) $\frac{7}{10} < \frac{3}{4}$

78) $\frac{1}{3} < \frac{10}{12}$

79) $\frac{9}{12} < \frac{4}{5}$

80) $\frac{7}{8} > \frac{7}{12}$

81) $\frac{3}{10} < \frac{2}{3}$

82) $\frac{4}{6} < \frac{9}{10}$

83) $\frac{10}{12} > \frac{1}{8}$

84) $\frac{4}{12} < \frac{3}{8}$

85) $\frac{5}{6} > \frac{4}{8}$

86) $\frac{4}{6} > \frac{2}{4}$

87) $\frac{6}{8} > \frac{2}{3}$

Comparing Unlike Fractions: Practice (continued)

Part 8 Compare (continued)

88) $\frac{1}{3}$ [>] $\frac{2}{8}$

89) $\frac{8}{10}$ [>] $\frac{1}{8}$

90) $\frac{6}{12}$ [=] $\frac{2}{4}$

91) $\frac{2}{3}$ [=] $\frac{4}{6}$

92) $\frac{1}{5}$ [<] $\frac{2}{3}$

93) $\frac{5}{6}$ [>] $\frac{2}{4}$

Part 9 Remember? Problems from earlier packets.

94) $\frac{9}{10} = \frac{[90]}{100}$

95) $56 \times 12 =$ [672]

96) $1{,}515 \div 5 =$ [303]

97) 4,816 to the nearest ten = [4,820]

98) $\frac{2}{3}$ of 27 = [18]

99) $8{,}003 - 2{,}457 =$ [5,546]

Comparing Unlike Fractions: Practice (continued)

Part 10 Word problems Draw strips to compare.

100) Mia ran 3/4 of a mile. Ben ran 2/3 of a mile. Who ran farther?

Number sentence **3/4 = 9/12, 2/3 = 8/12**

Answer **Mia**

Draw or show your work

101) A red jug is 3/8 full. A blue jug is 5/6 full. Which jug has more?

Number sentence **3/8 < 1/2 < 5/6**

Answer **blue** jug

Draw or show your work

102) Kim ate 4/10 of a pizza. Leo ate 2/5 of the same pizza. Who ate more?

Number sentence **4/10 = 2/5**

Answer **same**

Draw or show your work

103) Sam's ribbon is 7/12 of a yard. Jo's ribbon is 1/2 of a yard. Whose ribbon is longer?

Number sentence **7/12 > 6/12**

Answer **Sam's** ribbon

Draw or show your work

104) Challenge: Ana did 5/6 of her homework. Max did 7/8 of his. Who has less left to do?

Number sentence **1/8 left < 1/6 left**

Answer **Max**

Draw or show your work

Check-In: Comparing Unlike Fractions

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 More or less than half?

1) $\frac{11}{12}$ less than 1/2 = 1/2 (more than 1/2)

2) $\frac{3}{6}$ less than 1/2 (= 1/2) more than 1/2

Part 2 Make the bottoms the same

3) $\frac{3}{4} = \frac{9}{12}$ and $\frac{4}{6} = \frac{8}{12}$ so $\frac{3}{4}$ [>] $\frac{4}{6}$

4) $\frac{3}{4} = \frac{9}{12}$ and $\frac{5}{6} = \frac{10}{12}$ so $\frac{3}{4}$ [<] $\frac{5}{6}$

Part 3 Compare Write <, > or =.

5) $\frac{1}{4}$ [<] $\frac{7}{10}$

6) $\frac{5}{6}$ [>] $\frac{4}{12}$

7) $\frac{1}{8}$ [<] $\frac{6}{10}$

8) $\frac{8}{10}$ [>] $\frac{3}{12}$

9) $\frac{1}{8}$ [<] $\frac{8}{10}$

10) $\frac{2}{3}$ [>] $\frac{3}{12}$